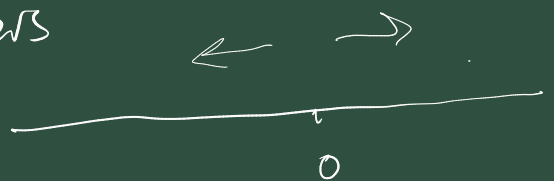


Sequences

$\{x_n\}_{n=1}^{\infty}$ the infinite sequence
of elements $x_1, x_2, x_3, \dots$

$\mathbb{Z} = \{ \dots -3, -2, -1, 0, 1, 2, 3, \dots \}$
the integers.

$\mathbb{R}$, the real numbers



$\{a_j\}_{j=1}^{\infty}$, $a_j = 2^{j-1}$

$P_n = \dots ?$

$\{s_n\}_{n=1}^{\infty} = \frac{1}{1}, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \dots$

4. Are there arithmetic geometric sequences?

* The ratio of neighboring arithmetic terms will be.

$$\frac{a + (n-1)d}{a + (n-2)d}$$

If this is constant for all n , what conditions does that impose?

if $d=0$

$$\frac{a+d}{a}, \frac{a+2d}{a+d}, \frac{a+3d}{a+2d}, \dots$$

||

$$\frac{a}{a}, \frac{a}{a}, \frac{a}{a}, \dots$$

So constant sequences,

($a \neq 0$)

$a, a, a, a, \dots$

will be arithmetic ($d=0$) and

geometric ($r=1$).

$$x_n, \quad x_n = \frac{n^2 + 2}{2^n}, \quad \text{for } n \geq 1$$

Ex 1.2.2.

Prove the increasing/decreasing behaviour of $\{a_m\}_{m=1}^{\infty}$

$$a_m = \frac{5^m}{m!}$$

$$(m! = m(m-1)(m-2) \dots 3 \cdot 2 \cdot 1)$$

Consider the ratio of neighboring terms

$$\frac{a_{m+1}}{a_m} = \frac{5^{m+1}}{(m+1)!} \cdot \frac{m!}{5^m}$$

$$= \frac{5}{m+1}$$

Compare this to 1.

$$\left(\frac{a_{m+1}}{a_m} = \frac{5}{m+1} \right) \begin{cases} > 1 & \text{for } m=1, 2, 3 \\ = 1 & \text{for } m=4 \\ < 1 & \text{for } m \geq 5 \end{cases}$$

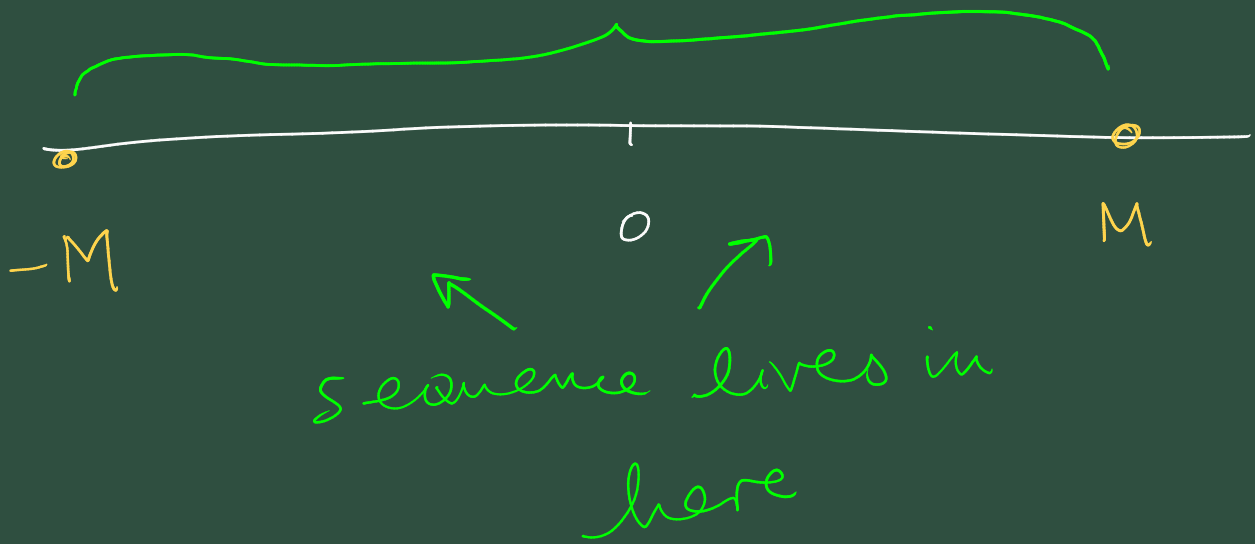
$$\text{So } a_{m+1} > a_m \quad \text{for } m=1, 2, 3$$

$$a_{m+1} = a_m \quad \text{for } m=4$$

$$a_{m+1} < a_m \quad \text{for } m \geq 5$$

So the sequence is eventually decreasing.

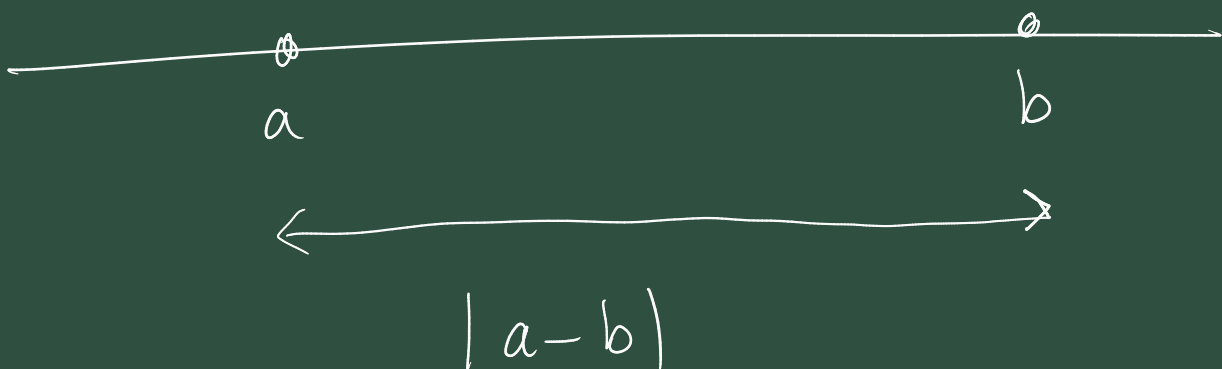
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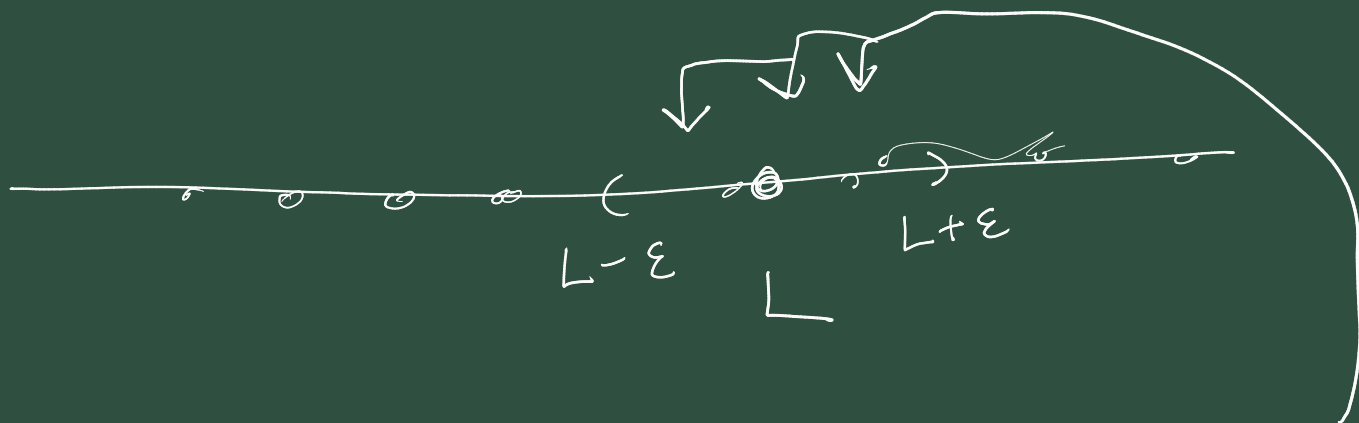


Consider the long-term behaviour

$$\text{of } \{x_n\}_{n=1}^{\infty}$$

$$x_n = \frac{n^2 + 2}{2^n}$$





$|x_n - L| < \epsilon$ means x_n lives

$\forall$ "for all"

$\exists$ "there exists"

ϵ "epsilon"

$\mathbb{N}$

Use the formal definition to prove that $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$.

Proof

Let $\epsilon > 0$ be given, we have to find the appropriate N_ϵ .

Consider $|x_n - L| < \varepsilon$

$$\Leftrightarrow \left| \frac{1}{n} - 0 \right| < \varepsilon$$

$$\Leftrightarrow \left| \frac{1}{n} \right| < \varepsilon$$

$$\Leftrightarrow \frac{1}{n} < \varepsilon$$

$$\Leftrightarrow n > \frac{1}{\varepsilon}$$

So let's ^{choose} N to be any integer greater than $\frac{1}{\varepsilon}$

and now, whenever

$$n \geq N$$

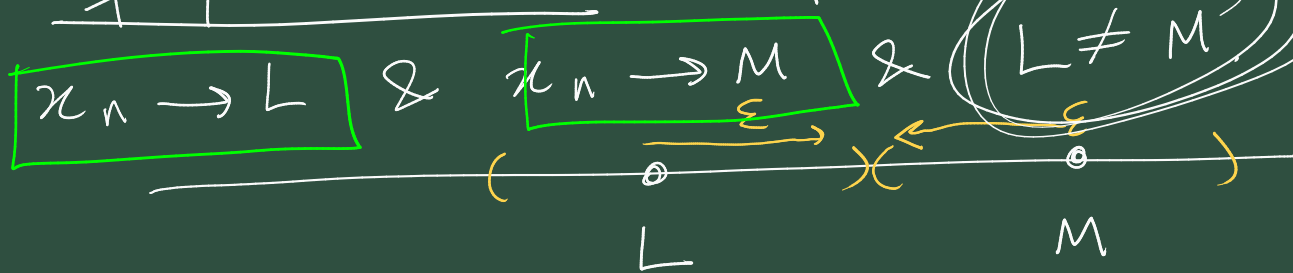
we will have $n \geq N > \frac{1}{\varepsilon}$

$$\Rightarrow \left| \frac{1}{n} - 0 \right| < \varepsilon$$

This proves that $\frac{1}{n} \rightarrow 0$
as $n \rightarrow \infty$.

Proof of theorem 1.4.2

Assume



Use the value

$$\varepsilon = \frac{|L - M|}{2}$$

Since $x_n \rightarrow L$ there exists N_1
such that for all $\underline{n \geq N_1}$ $|x_n - L| < \varepsilon$

And since $x_n \rightarrow M$ there exists N_2
such that for all $\underline{n \geq N_2}$ $|x_n - M| < \varepsilon$

Set $N = \max(N_1, N_2)$

So for all $n \geq N$ $|x_n - L| < \varepsilon$
& $|x_n - M| < \varepsilon$

So for $n \geq N$

$$|L - M| = |L + x_n - x_n - M|$$

$$= |x_n - M + (-x_n + L)|$$

$$\leq |x_n - M| + |-x_n + L|$$

$$= |x_n - M| + |x_n - L|$$

$$< \frac{|L - M|}{2} + \frac{|L - M|}{2}$$

$$= |L - M|$$

This is a contradiction.

So therefore $L = M$.